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Short communication

Effect of corruption on healthcare satisfaction in post-soviet nations: A cross-country instrumental variable analysis of twelve countries

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ABSTRACT

There is the lack of consensus about the effect of corruption on healthcare satisfaction in transitional countries. Interpreting the burgeoning literature on this topic has proven difficult due to reverse causality and omitted variable bias. In this study, the effect of corruption on healthcare satisfaction is investigated in a set of 12 Post-Socialist countries using instrumental variable regression on the sample of 2010 Life in Transition survey (N = 8655). The results indicate that experiencing corruption significantly reduces healthcare satisfaction.

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1. Introduction

The purpose of this paper is to explore the effect of corruption on healthcare satisfaction in Post-Socialist countries. On one end of the equation, customer satisfaction with healthcare is recognized as a crucial component of healthcare delivery by governments, healthcare authorities practitioners, and patients worldwide (Smith et al., 2006; Kimenyi and Shughart II, 2006; Kettl et al., 2006; Amponsah-Nketiah and Hiemenz, 2009). Feedback from customers provides an important impetus to improving healthcare delivery (Qatari and Haran, 1999; Bara et al., 2002; Brinkerhoff and Wetterberg, 2013). Satisfied customers are more likely to develop long-lasting relationship with healthcare providers and demonstrate higher level of compliance, which ultimately leads to better health outcomes (Margolis et al., 2003; Bleich et al., 2009; Njong and Tchouapi, 2014).

On the other side of equation, corruption in healthcare exists as a rampant issue in Post-Socialist countries (Bonilla-Chacin et al., 2005; Falkingham et al., 2010). The literature notably lacks consensus regarding the effect of corruption in developing and transitional countries in general, and in healthcare in particular. One school of thought conceptualises corruption as "sand the wheels" and suggests a negative effect of corruption (Clausen et al., 2011). Indeed, previous studies on healthcare in transitional countries support this view. Corruption, encompassing unofficial out-of-pocket payments and gifts, is associated with lower propensity of using healthcare when needed (Balabanova et al., 2004; Falkingham, 2004; Fan and Habibov, 2009). Bribes often constitute catastrophic expenditures for the poor (Habibov, 2009a, 2011). Due to corruption barriers, more advanced and specialized health services remain out of reach for the poor (Habibov, 2009b, 2010). Conceptualizing corruption as "sand the wheel" postulates that we should expect the effect of corruption on healthcare satisfaction.

The opposite school of thought conceptualises corruption as "grease in the wheels" and highlights the positive outcomes of corruption (Méon and Weill, 2010). First, corruption alleviates inefficiencies of administering public healthcare in transitional period. Healthcare professionals consider their remuneration low and expect informal payments, while patients expect that they would have to pay out-of-pocket to underpaid professionals for additional or better quality services (Gaal and McKee, 2004; Vian and Burak, 2006). When expectations of healthcare professionals and patients match, then a transaction of paying and receiving unofficial payments takes place. Corruption also introduces redistribution towards the poor. A number of previous studies report that healthcare professionals charge a lower out-of-pocket rate or even provide free care to citizens struggling with poverty,

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compensating the "lost" revenue by asking wealthier patients for higher payments (Ensor and Savelyeva, 1998; Belli et al., 2004; Gotsadze et al., 2005). In addition, corruption encourages competition. Individuals may pay bribes to receive necessary treatment "free" in public healthcare rather than to pay officially more for the same treatment in private facilities (Rose, 1998). Conceptualizing corruption as "grease the wheels" postulates that we should expect effect of healthcare satisfaction on corruption.

Yet, another school of thought considers corruption a cultural norm (Turex, 2011; Wang-Sheng and Guven, 2013). Thus, although large-scale corruption schemes are commonly denounced, out-of-pocket payments to healthcare professionals are not considered an act of corruption (Bowser, 2001). Conceptualising corruption as a harmless cultural norm suggests that there is no statistically significant association between corruption and satisfaction in either direction.

Given the lack of consensus about the effect of corruption on healthcare satisfaction, we focus on testing the above-describe three hypotheses on a diverse sample of 12 Post-block countries. We use classic single-stage linear OLS regression to test whether healthcare corruption is a cultural norm, and examine if it has a statistically significant link with satisfaction. If OLS identifies such a link, then we can reject conceptualization of corruption as a cultural norm. However, results of single-stage models like OLS are prone to endogeneity, namely, reverse causality and omitted variable bias. OLS cannot rule out reverse causality. For instance, it is plausible to assume that clients satisfied with the higher level of healthcare service pay extra or give gifts to healthcare personnel, supporting "grease the wheels" conceptualization (Rose, 1998; Gaal and McKee, 2004; Vian and Burak, 2006; Habibov, 2010). It is also plausible that OLS estimation can suffer from uncontrolled confounder variable that affect both corruption and satisfaction leading to omitted variable bias. Consequently, we estimate and rigorously test two-stage 2SLS regression which addresses both reverse causality and omitted variable bias (Wooldridge, 2008). If 2SLS reports negative association between corruption and satisfaction, then we can reject the "grease the wheels" conceptualization of corruption in favour of the "sands the wheels". An excellent discussion of employing 2SLS to address reverse causality and omitted variable bias in healthcare studies is provided by Kim et al. (2011). To the best of our knowledge, our study is the first one to apply 2SLS to test three different conceptualisations of corruption and to establish causal association between corruption and healthcare satisfaction.

Let us now turn to method section.

2. Method

We used data from the Life-In-Transition survey (the LITS), which was conducted in 2011 by European Bank of Reconstruction and Development in cooperation with the World Bank (EBRD, 2009; Habibov, 2013). Our sample covers Post-Socialist Armenia, Azerbaijan, Belarus, Georgia, Kazakhstan, Kyrgyzstan, Moldova, Mongolia, Russia, Tajikistan, Ukraine, and Uzbekistan. Approximately 1000 respondents in non-institutionalized populations were interviewed face-by-face in each country by especially trained interviewers. However, since our focus is on healthcare satisfaction, our sample is limited to the respondents who reported using public healthcare within the last 12 months.

The LITS uses a clustered sampling design. According to LITS manual (Ipsos, 2011), communities are clusters with clearly defined borders (e.g. census enumeration areas or voting districts) based on the most recent national censuses or election lists (CITE). Clusters are selected for the survey based on the probability proportional to size. Each country included approximately 50–70 clusters depending on the geographical and population size. Within the clusters, a random walk fieldwork method was employed select a household for the interview. Maximum of 20 households were selected randomly for the interview in each cluster. Finally, within the households, the "last birthday" method was employed to select the respondent for the interview. Replacement was not allowed to avoid selection bias. The Research Ethics Board of University of Windsor does not require ethical approval for secondary data analysis.

2.1. Outcome, predictor, and controls

Our outcome variable is satisfaction with healthcare. The LITS asked respondents to rate their satisfaction with the quality and efficiency of the public healthcare system. Satisfaction is measured on five-point ordinal scale, ranging from "very unsatisfied" to "very satisfied". Satisfaction is treated as continuous measure across all estimations (Kim et al., 2011). Our predictor variable is corruption in healthcare. The LITS asked whether an unofficial payment was made or gift was given to public healthcare personnel in the last 12 months. The responses are binomial (Yes = 1, No = 0).

Socio-demographics are controlled by age, gender, and education of respondent. Healthcare quality is controlled by the index of healthcare quality. The LITS asked respondents a set of question regarding problems they may have encountered in healthcare, such as frequent and unjustified absence of doctors, treated disrespectfully by personnel, availability of medication, long waiting times, and unclean facilities. The responses are binomial (no such a problem = 1, otherwise = 0). Summing up these binomial answers, we created an additive index varying from 0 to 5, where a higher index value represents higher quality of healthcare received. To control for needs for healthcare, we use a binomial variable of poor health status, where value of 1 denotes a respondent who reported poor or very poor health. Tertiles of wealth index represent middle 33.3% and wealthiest 33.3% of households in each country (Filmer and Pritchett, 2001). Descriptive statistics is reported in Table 1.

2.2. Analysis

We commence by using OLS, where the outcome variable is satisfaction with healthcare, the predictor variable is corruption in healthcare with individual-level controls are as discussed above. Next, we estimate a 2SLS model. In the first-stage, corruption is regressed on two instruments and covariates, including fixed effects. In the second-stage, healthcare satisfaction is regressed on the predicted value of corruption from the first stage and

Table 1
Descriptive statistics.

	Mean	Std. Dev.	Min	Max	Yes %)
Outcome					
Satisfaction with healthcare	3.252	1.066	1	5	
Predictor					
Experienced corruption in healthcare		0.005	0	1	37%
Controls					
Healthcare service quality	3.959	1.202	0	5	
Poor health		0.37	0	1	16%
Age	43.658	17.024	17	93	
Female		0.005	0	1	66%
University		0.005	0	1	25%
Middle wealth households		0.005	0	1	30%
Wealthiest households		0.005	0	1	31%
Work		0.005	0	1	49%
Instruments					
Ask for interference		0.004	0	1	12%
Frequency of public service utilization	2.201	1.155	1	8	

covariates, including fixed effects. Two instruments, "Ask for interference" and "Frequency of public service utilization", are used in the 2SLS model. Usage of these instruments is supported by both theoretical and empirical considerations.

The existing literature suggest that individuals predisposed to bending rules have less ethical impediment to corruption, even having a higher propensity to accept and be involved in corruption in general (Lavallée et al., 2008; Cho and Kirwin, 2007). The LITS asks how likely respondents are to ask friends, relatives, and classmates to help to resolve various problems, such as receiving permits and other official documents or receiving acceptance to university. The responses of "likely" and "very likely" were added to create the binomial instrument "Ask for interference," where value of 1 denotes higher propensity for breaking the rules by an individual. The existing literature suggests that toleration of corruption is associated with frequency of public service unitization (Grosjean et al., 2013). The LITS asks whether respondents utilized various public services, for instance, public primary and secondary education, public vocational education, social services, the courts for civil matters, and road police for the last 12 months. Positive responses were tallied to compute the continuous instrument "Frequency of public service utilization," where the value of the variable varies between 0 and 8, where a higher number represents more frequent public service utilization.

The estimation of 2SLS requires instruments, which must be correlated with a predictor. To test this assumption, we estimate the first stage F statistic that is 34.41, $p = 0.000$. The F statistics is significant and higher than 10, signalling that jointly our instruments are correlated with the predictor (Stock and Yogo, 2002). Likewise, minimum eigenvalue statistic is 37.07, which is higher than Stock and Yogo's critical values of 19.93, suggesting that our instruments are not weak (Stock et al., 2002; Stock and Yogo, 2005). Since we have two instruments for one predictor, we used a likelihood ratio test to assess redundancy of each instrument (Baum, 2006). The results of the test are consistently significant, suggesting that both instruments provide useful information to identify the first stage equation.

At the same time, the instruments should not be correlated with the outcome variable other than through their effect on the predictor. To test this assumption, we perform Wooldridge's (1995) score test of overidentifying restrictions that is analog of Sargan test for clustered data. Non-significant results of these tests suggest that the instruments are jointly uncorrelated with the outcome variable. We also compute Pearson correlation between each instrument and outcome variable. The correlation between each instrument individually and outcome variable is negligible, specifically, $r = -0.04$ for "Ask for interference" instrument and $r = -0.08$ for "Frequency of public service utilization" instrument. In addition, C statistics for each instrument in both models is not significant, suggesting that each of the instruments individually is exogenous (Baum, 2006).

Finally, to test the endogeneity, we estimated Wooldridge's (1995) regression-based tests of endogeneity which are analog for Durbin–Wu-Hausman test for clustered data. Significant results of the test confirms that experienced corruption index is endogenous. Consequently, 2SLS should be estimated instead of single-stage OLS.

2.3. Findings

To better understand the relation between corruption and healthcare satisfaction we plot them against each other in Fig. 1. As observed, corruption correlated with lower level of satisfaction. The visual correlation is supported by the Pearson correlation ($r = -0.53$) and Spearman correlation ($r = -0.52$).

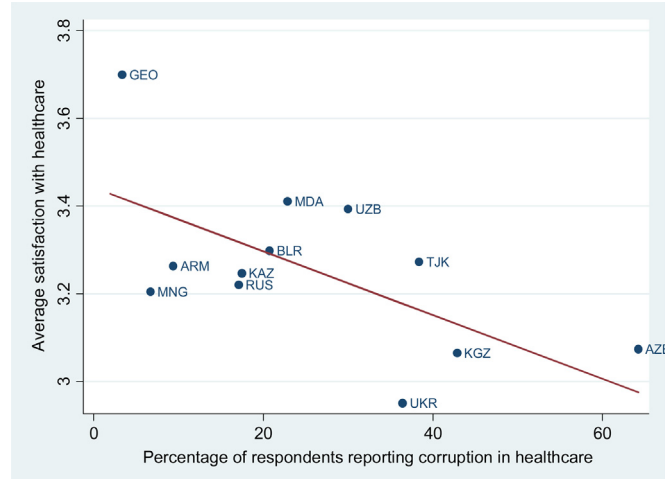

Fig. 1. Plot of satisfaction with healthcare against percentage of respondents reporting corruption.

The results of one stage OLS are reported in first column of Table 2. Corruption in public healthcare leads to significant reduction in satisfaction by a factor of −0.46. Reporting poorer health status is associated with reduction in satisfaction, while an improvement in quality of healthcare has opposite effect. Other covariates are not significant.

The results of the main stage of 2SLS with instruments "Ask for interference" and "Frequency of public service utilization" are reported in Model 2 in Panel A of Table 2. After taking into account endogeneity, corruption has a negative effect on health satisfaction by a factor of −1.26. The results for the covariates are very similar to the results of OLS without any systematic differences in sign and magnitude. Comparing 2SLS and OLS, we can see that the simple one-stage OLS has considerably underestimated the negative effect of corruption on satisfaction. Once endogeneity is taken into account by the 2SLS, the magnitude of the negative corruption effect considerably increases. The 2SLS indicates a negative effect by a factor of −1.26, while OLS indicates a negative effect by a factor of −0.46, which translates to approximately 2.7 times the difference between the estimates. This finding suggests a strong endogeneity bias in corruption.

Another way to interpret results of 2SLS is to compare the coefficient for corruption effect with the raw mean for satisfaction of 3.25 which is reported in Table 1. The results of model 2 indicate that experiencing corruption reduces satisfaction by approximately 38% of the raw mean, while the results of model 3 suggest that corruption reduces satisfaction by approximately 36% of the raw mean.

The result of the first stage of 2SLS is reported in Panel B of Table 2. All instruments are significant and correlated with the endogenous variable of interest in the expected direction. Thus, as it could be expected asking for interference and frequency of public service utilization are associated with more corruption.

2.4. Sensitivity analysis

We test sensitivity of the 2SLS models in two ways. First, we test whether healthcare quality may have mediating, rather than moderating relationship between making unofficial payments and satisfaction with healthcare, since individuals who make such payments may receive higher quality of care and ultimately report greater satisfaction. Hence, we re-estimated models 2 without healthcare quality. The results are reported in model 3. As shown,

Table 2
Results of regression analysis.

	Model 1		Model 2		Model 3		Model 4	
	OLS		2SLS		2SLS		2SLS	
	Coeff.	p-value	Coeff.	p-value	Coeff.	p-value	Coeff.	p-value
Panel A: Results of OLS and main stage of 2SLS								
Experienced corruption in healthcare	−0.461	0.000	−1.266	0.000	−1.820	0.000	−1.220	0.000
	0.026)		0.280)		0.239)		0.277)	
Healthcare service quality	0.278	0.000	0.207	0.000			0.211	0.000
	0.010)		0.027)				0.027)	
Poor health	−0.219	0.000	−0.202	0.000	−0.239	0.000	−0.206	0.000
	0.033)		0.035)		0.040)		0.035)	
Age	0.001	0.069	0.000	0.856	0.000	0.808	0.000	0.937
	0.001)		0.001)		0.001)		0.001)	
Women	0.019	0.391	0.009	0.690	0.002	0.952	0.008	0.743
	0.022)		0.024)		0.026)		0.024)	
University	−0.005	0.842	0.003	0.908	−0.010	0.734	0.011	0.67
	0.026)		0.027)		0.031)		0.026)	
Middle wealth households	−0.037	0.149	−0.036	0.197	−0.046	0.139		
	0.026)		0.028)		0.031)			
Wealthiest households	0.009	0.731	0.022	0.447	0.008	0.816		
	0.027)		0.029)		0.032)			
Work							−0.014	0.567
							0.024)	
Community dummies included	Yes		Yes		Yes		Yes	
Number of observations	8494		8494		8494		8494	
F statistics	28.5	0.000						
Wald chi2			2026.54	0.000	763.20	0.000	2054.65	0.000
R-squared	0.22		0.12		N.A.		0.13	
First-stage regression summary statistics								
Robust F			34.41	0.000	55.21	0.000	34.72	0.000
Minimum eigenvalue statistic			37.07		59.56		37.26	
Stock and Yogo's critical value			19.93		19.93		19.93	
LM test for redundancy of Ask for interference instrument			14.50	0.000	17.46	0.000	14.37	0.000
LM test for redundancy of Frequency of public service utilization instrument			56.23	0.000	95.96	0.000	56.85	0.000
Test of overidentifying restrictions:								
Score chi2			0.02	0.875	0.11	0.739	0.05	0.829
C statistic for of Ask for interference instrument			0.03	0.869	0.12	0.726	0.05	0.821
C statistic for of Frequency of public service utilization instrument			0.03	0.869	0.12	0.726	0.05	0.821
Tests of endogeneity								
Robust score chi2			9.18	0.003	31.05	0.000	8.25	0.004
Robust regression F			9.14	0.003	31.26	0.000	8.21	0.004
Panel B: Result of first-stage of 2SLS								
Ask for interference			0.054	0.000	0.061	0.000	0.054	0.000
			0.015)		0.016)		0.015)	
Frequency of public service utilization			0.032	0.000	0.043	0.000	0.032	0.000
			0.004)		0.005)		0.004)	
Healthcare service quality			−0.084	0.000			−0.084	0.000
			0.004)				0.004)	
Poor health			0.025	0.070	0.052	0.000	0.028	0.046
			0.014)		0.014)		0.014)	
Age			−0.001	0.002	−0.001	0.000	−0.001	0.005
			0.000)		0.000)		0.000)	
Women			−0.010	0.317	−0.009	0.398	−0.007	0.484
			0.010)		0.010)		0.010)	
University			0.009	0.440	0.018	0.113	0.007	0.550
			0.011)		0.012)		0.011)	
Middle wealth households			−0.003	0.818	0.001	0.900		
			0.012)		0.012)			
Wealthiest households			0.001	0.945	0.008	0.533		
			0.012)		0.013)			
Work							0.017	0.085
							0.010)	
Community dummies included			Yes		Yes		Yes	
Number of observations			8494		8494		8494	
F statistics			40.70	0.000	32.74	0.000	41.12	0.000
R-squared			0.24		0.20		0.24	

Note. Robust standard errors are in brackets.

the direction and significance of corruption effect in models 2 is similar to those in models 3. The magnitude of the effect, however, is higher when we drop healthcare quality. For instance, effect of corruption without taking into consideration effect of quality in model 3 is approximately 43% more negative than in model 2 with healthcare quality as a control. It appears that healthcare quality plays moderating rather than mediating role between making unofficial payments and satisfaction with healthcare and hence should be controlled for.

Second, the question could be asked whether individuals in the

wealthiest tertile in a poor country (e.g. Tajikistan) are actually richer than individuals in the second or even the last tertile in a wealthier country (e.g. Hungary). Consequently, we test weather changing welfare tertiles for a being employed may significantly alter the results. The binomial variable employed has value of 1 if respondents reported to work for income in last 12 months. We re-estimated models 2 with employment status instead of tertiles of wealth. The results reported in models 4. Again, the direction and significance are similar to our original 2SLS models. The negative effect of corruption in models 2 is −1.26 which is similar to −1.22 in model 4.

All other covariates in sensitivity analysis models are similar to those in models 2, 3, and 4. Importantly, all models in sensitivity analysis passed all required instrumental variable tests.

2.5. Simulated effect of corruption

Based on the 2SLS results in models 2, we can simulate the effect of corruption on satisfaction with healthcare by predicting the average satisfaction for every individual in the sample depending on corruption. The results of simulation are reported in Table 3. The first row of the table shows the average raw mean of 3.25. The next row shows the mean of satisfaction assuming every respondent reported experiencing corruption. Experiencing corruption by everyone would reduce mean satisfaction to 2.45, which constitutes 76% of the raw mean. By contrast, the last row shows the mean assuming no respondents reported experiencing corruption. Having no reported corruption would boost mean satisfaction to 3.72, which is 14% higher than the raw mean.

2.6. Discussion, implications, and limitations

There is the lack of consensus about the effect of corruption on healthcare satisfaction in transitional countries. Interpreting the burgeoning literature on this topic is difficult due to reverse causality and omitted variable bias. In this study, we test "grease the wheels", "sand in the wheels", and "cultural norm" hypotheses about possible effect of corruption on healthcare satisfaction.

The results of OLS suggest that corruption is associated with significant reduction in healthcare satisfaction. The result of 2SLS, which was estimated to address reverse causality and omitted variable bias, provides qualitatively similar results. Likewise, simulation results demonstrate that reducing corruption will considerably increase healthcare satisfaction. Importantly, these results are stable when controlled for alternative explanations of healthcare satisfaction such as, quality of care received, and health and wealth statuses.

These findings have both theoretical and practical significance. From a theoretical perspective, our findings reject the notion of corruption as "grease in the wheels" and a "cultural norm" in favour of the "sand in the wheels" hypothesis inasmuch as we found significantly negative effects of corruption on satisfaction. From practical perspective, our findings highlight the importance of reducing corruption in healthcare. One way to reduce corruption is to legitimize informal payments in the framework of broader reforms of healthcare sector in transitional countries. The different approaches to such legitimizations are currently being discussed (e.g. Gaal and McKee, 2004; Falkingham et al., 2010; WHO, 2015). Another method is a step-up enforcement and to enhance transparency by joining international initiatives in healthcare reform (Vian, 2008). In any case, given the strong negative effect of corruption on satisfaction identified by this study, the public health gains from containing and reducing corruption is to be significant.

Our study is not without limitations. First, since we used secondary data survey, our definition of corruption is limited to petty corruption, rather than large-scale corruption schemes. At the same time, measures of corruption and healthcare satisfaction were not validated across countries. Besides, corruption is viewed from the customer perspective rather that from the perspective of other agents, for instance, healthcare personnel and government regulators. Likewise, we could not differentiate between beneficiaries of corruption among healthcare personnel and reasons for paying bribes.

In addition, although we found no empirical evidence for direct effect of the instrumental variables on outcome variable in our main model and sensitivity analysis models, nevertheless, such direct effect of our instruments could not be ruled out completely. Finally, the omitted community-level factors, for instance, community-level spending per capita on education, infrastructure, and indeed on health care, may vary over time and be potentially correlated with the utilization of public services. Equally, we cannot control for omitted individual-level factors which may also vary across time.

Table 3
Results of simulation.

Scenario	Mean of satisfaction with healthcare
Raw sample mean	3.25***
	0.007)
Everyone experienced corruption	2.45***
	0.003)
No one experienced corruption	3.72***
	0.003)

Note. Bootstrapped standard errors are in parenthesis. *p < 0.05, **p < 0.01, ***p < 0.001.

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